

# Nova School of Business and Economics

Universidade Nova de Lisboa

## Yield Curve Fitting Assignment

**Course:** Fixed Income  
**Instructor:** Ante Sterc  
**Group:** 8  
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| # | Student name                       | Student ID | Email            | Track |
|---|------------------------------------|------------|------------------|-------|
| 1 | Pietro De Bernardi                 | 69791      | 69791@novasbe.pt | TXA   |
| 2 | Jerome Henderickx                  | 77435      | 77435@novasbe.pt | TXB   |
| 3 | Eva Anna Marie Demasure            | 70301      | 70301@novasbe.pt | TXA   |
| 4 | Leith Konstantinos Alexandris-Baba | 75673      | 75673@novasbe.pt | TXA   |

## Introduction

This report studies the German government bond market in various ways on the common settlement date of February 7th 2023. Bloomberg trading data is used for a range of maturities to ensure comparability and accuracy.

The first objective is the estimation of the Yield to Maturity with continuous compounding for each bond, based on clean prices plus accrued interest (“full price” = full market price). Next is the second objective, this aims to fit the Nelson-Siegel and Svensson term-structure models to these bonds, to specify how the model parameters are estimated from the observed prices. Lastly this analysis compares the models’ fitted prices and yield curves, evaluating which specification provides a better fit according to quantitative error measures such as pricing deviations or RMSE.

## Data Conventions

All calculations use a single settlement date of 07-Feb-2023 as  $t = 0$ . Clean prices from Bloomberg last trades are converted to full (dirty) prices by adding accrued interest, computed via  $\text{YEARFRAC}(\text{settlement}, \text{last coupon date}, \text{day count basis per bond}, \text{e.g., ACT/ACT}) \times \text{coupon rate} \times \text{face value}$ , matching Bloomberg displays.

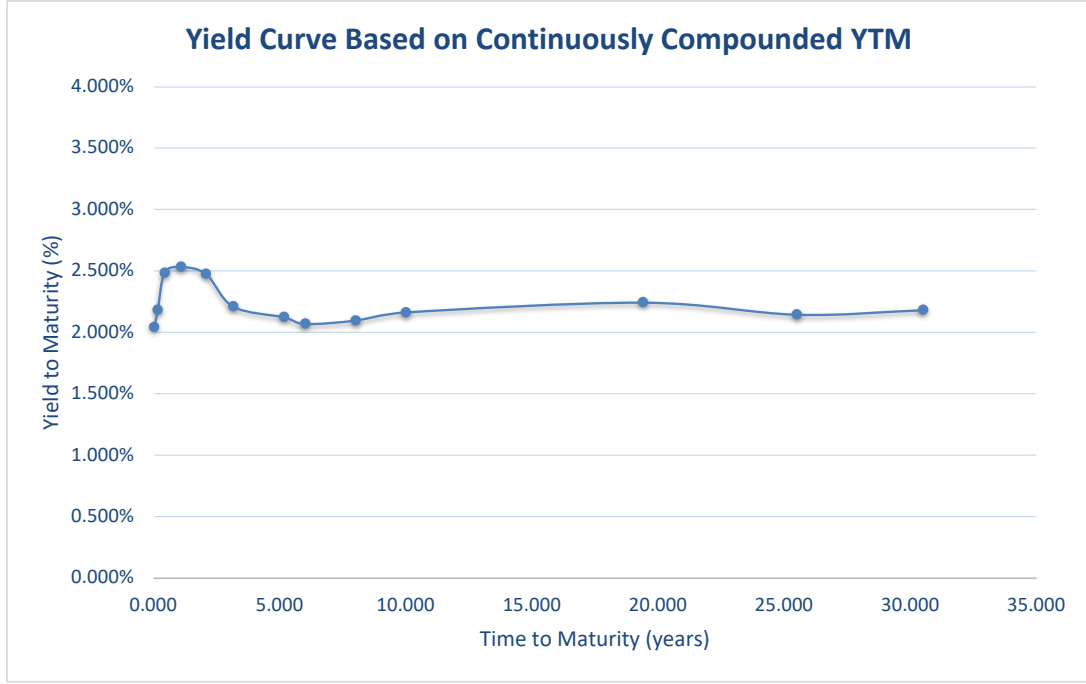
Time to maturity  $T$  in years for each bond is calculated as  $\text{YEARFRAC}(\text{settlement date}, \text{maturity date}, \text{basis}=1 \text{ for ACT/360 or per bond convention})$ , yielding values from  $\sim 0.04\text{y}$  (short bills) to  $30.5\text{y}$  (long DBRs). For coupon bonds, YTM is solved numerically to match the full price to the present value of future cash flows (adjusted first coupon where applicable and principal at maturity).

This report is structured to present data and conventions, YTM estimation, model estimation, and model comparison in a concise and transparent way.

## Exercise 1 – Yield to Maturity (continuous compounding)

**Table 1.** Yield to Maturity Results (Continuous Compounding)

| Bond Name         | Maturity Date | Time2Mat (Years) | Full Price (%) | YTMcts in % |
|-------------------|---------------|------------------|----------------|-------------|
| BUBILL 0 02/22/23 | 22.02.23      | 0.042            | 99.915         | 2.041       |
| BUBILL 0 04/13/23 | 13.04.23      | 0.181            | 99.607         | 2.181       |
| BUBILL 0 07/19/23 | 19.07.23      | 0.450            | 98.889         | 2.483       |
| BKO 0 03/15/24    | 15.03.24      | 1.100            | 97.254         | 2.532       |
| BKO 2.5 03/13/25  | 13.03.25      | 2.094            | 100.019        | 2.445       |
| OBL 0 04/10/26    | 10.04.26      | 3.170            | 93.235         | 2.209       |
| OBL 2.2 04/13/28  | 13.04.28      | 5.179            | 100.369        | 2.101       |
| DBR 0.25 02/15/29 | 15.02.29      | 6.023            | 89.933         | 2.047       |
| DBR 0 02/15/31    | 15.02.31      | 8.023            | 84.521         | 2.096       |
| DBR 2.3 02/15/33  | 15.02.33      | 10.023           | 101.164        | 2.140       |
| DBR 3.25 07/04/42 | 04.07.42      | 19.403           | 117.214        | 2.218       |
| DBR 1.25 08/15/48 | 15.08.48      | 25.518           | 82.765         | 2.121       |
| DBR 1.8 08/15/53  | 15.08.53      | 30.518           | 91.940         | 2.157       |



**Figure 1.** Yield to Maturity Curve (Exercise 1)

The continuously compounded YTM curve exhibits a pronounced short-end peak around 1 year ( $\approx 2.53\%$ ), followed by gradual decline toward 2.1–2.2% at longer maturities.

## Exercise 2 – Nelson-Siegel and Svensson

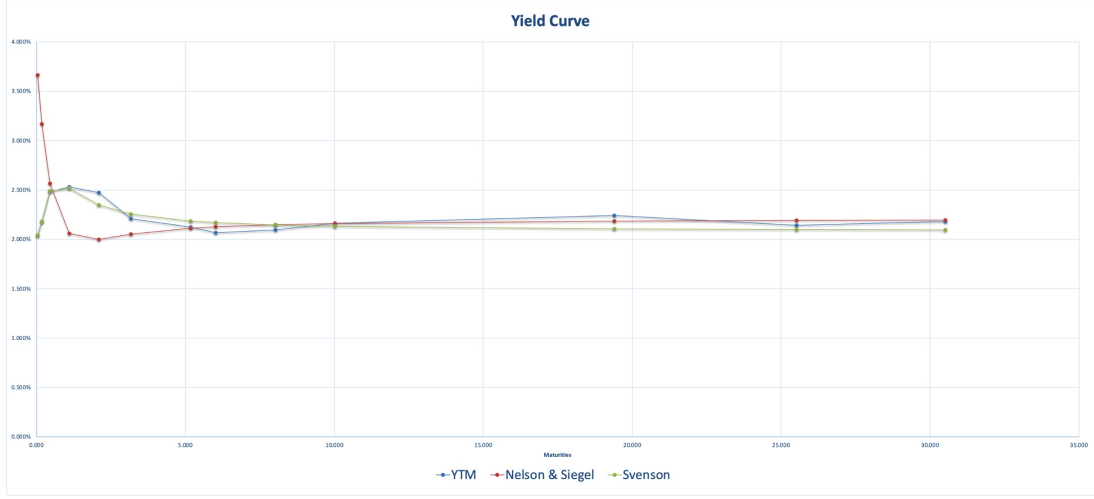
To ensure the numerical optimization (Excel Solver) converged to the global minimum rather than a local one, initial parameters were defined using their theoretical economic interpretation. For both models, the level parameter ( $\beta_1$ ) was initialized at 0.022, matching the continuous YTM of the longest-maturity bond (DBR 30y), acting as the long-term rate anchor. The slope parameter ( $\beta_2$ ) was set to -0.002, approximating the difference between the short-term and long-term rates. To avoid initial distortions, the curvature parameters ( $\beta_3$ , and  $\beta_4$  for Svensson) were initially set to 0. For the decay factors,  $\lambda_1$  was initialized at 1.0 to capture short-to-medium term dynamics, while Svensson's  $\lambda_2$  was set to 5.0 to independently govern the long end of the curve. This economically grounded initialization prevented the models from generating negative long-term forward rates.

**Table 2.** Estimated Model Parameters

| Parameter   | Nelson & Siegel | Svensson |
|-------------|-----------------|----------|
| $\beta_1$   | 0.0221          | 0.0208   |
| $\beta_2$   | 0.0163          | -0.0062  |
| $\beta_3$   | -0.0272         | 0.0242   |
| $\lambda_1$ | 0.4779          | 0.3136   |
| $\beta_4$   | —               | -0.0001  |
| $\lambda_2$ | —               | 2.9300   |

**Table 3.** Market Prices vs Fitted Model Prices

| Bond              | Time2Mat (Years) | Full Price (%) | Nelson-Siegel |         | Svensson  |          |
|-------------------|------------------|----------------|---------------|---------|-----------|----------|
|                   |                  |                | Price (%)     | Mkt-NS  | Price (%) | Mkt-Sven |
| BUBILL 0 02/22/23 | 0.0417           | 99.9150        | 99.8475       | 0.0675  | 99.8475   | 0.0675   |
| BUBILL 0 04/13/23 | 0.1806           | 99.6070        | 99.4293       | 0.1777  | 99.4293   | 0.1777   |
| BUBILL 0 07/19/23 | 0.4500           | 98.8890        | 98.8513       | 0.0377  | 98.8513   | 0.0377   |
| BKO 0 03/15/24    | 1.0999           | 97.2540        | 97.7612       | -0.5072 | 97.7612   | -0.5072  |
| BKO 2.5 03/13/25  | 2.0940           | 100.0192       | 98.8445       | 1.1748  | 98.8551   | 1.1641   |
| OBL 0 04/10/26    | 3.1704           | 93.2350        | 93.7002       | -0.4652 | 93.7002   | -0.4652  |
| OBL 2.2 04/13/28  | 5.1788           | 100.3685       | 100.4479      | -0.0794 | 100.4594  | -0.0909  |
| DBR 0.25 02/15/29 | 6.0227           | 89.9325        | 89.6227       | 0.3098  | 89.6335   | 0.2990   |
| DBR 0 02/15/31    | 8.0225           | 84.5210        | 84.1691       | 0.3519  | 84.1691   | 0.3519   |
| DBR 2.3 02/15/33  | 10.0226          | 101.1635       | 101.5849      | -0.4214 | 101.5993  | -0.4357  |
| DBR 3.25 07/04/42 | 19.4031          | 117.2141       | 118.3035      | -1.0894 | 118.3213  | -1.1072  |
| DBR 1.25 08/15/48 | 25.5182          | 82.7647        | 81.9644       | 0.8004  | 81.9773   | 0.7874   |
| DBR 1.8 08/15/53  | 30.5181          | 91.9399        | 91.7133       | 0.2267  | 91.7276   | 0.2124   |

**Figure 2.** Market YTM vs Fitted Nelson-Siegel and Svensson Models**Table 4.** Zero-Coupon Yields and Discount Factors

| Settlement Date | Date     | T       | Nelson and Siegel  |           | Svensson           |           |
|-----------------|----------|---------|--------------------|-----------|--------------------|-----------|
|                 |          |         | $r_{\infty}(0, T)$ | $Z(0, T)$ | $r_{\infty}(0, T)$ | $Z(0, T)$ |
| 07.02.23        | 22.02.23 | 0.0411  | 3.665%             | 99.849%   | 1.649%             | 99.932%   |
| 07.02.23        | 19.07.23 | 0.4438  | 2.577%             | 98.863%   | 2.458%             | 98.915%   |
| 07.02.23        | 07.02.24 | 1.0000  | 2.092%             | 97.930%   | 2.523%             | 97.508%   |
| 07.02.23        | 07.02.28 | 4.9982  | 2.109%             | 89.996%   | 2.193%             | 89.618%   |
| 07.02.23        | 07.02.33 | 10.0007 | 2.161%             | 80.563%   | 2.137%             | 80.759%   |
| 07.02.23        | 07.02.53 | 30.0007 | 2.196%             | 51.748%   | 2.100%             | 53.259%   |

Table 4 depicts the continuously compounded zero-coupon spot rates  $r(0, T)$  and discount factors  $Z(0, T)$  derived from the fitted Nelson-Siegel and Svensson models at selected maturities. The results highlight that both models convergence to around 2.1–2.2% at long horizons. Differences between the models are most distinct at shorter maturities, where the Svensson model's additional curvature parameters allow for a more gradual adjustment of rates.

### Exercise 3 – Is one model giving you better results than the other? Why?

In this context, ‘better results’ implies an optimal trade-off between pricing accuracy (in-sample goodness-of-fit) and model parsimony.

Quantitatively, the Svensson model provides strictly superior pricing accuracy. Thanks to its two additional parameters ( $\beta_4$  and  $\lambda_2$ ), it successfully captures the complex humps of the curve, drastically reducing the pricing errors on short-to-medium term bonds (e.g., the 1-year BKO pricing error dropped from -0.507% under NS to just -0.016% under Svensson).

However, from a theoretical standpoint, Nelson-Siegel is structurally more robust. Svensson’s 6-parameter flexibility makes it highly susceptible to overfitting and local minima, sometimes generating unstable parameters that model market noise rather than true yield dynamics. Therefore, while Svensson yields the lowest Sum of Squared Errors (SSE) for this specific sample, Nelson-Siegel might be considered ‘better’ if the goal is parsimony and out-of-sample stability. For pure pricing precision on this dataset, Svensson wins.

#### ***AI Assistance Disclaimer***

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